

EduPulse AI — Project Documentation

Next.js 16 Adaptive Learning & AI Tutor Platform

Fullstack Developer Fulltime Assignment Solution • House of Edtech (Sep 2026)

1. Executive Summary & Problem Solved

EduPulse AI is a full-stack, enterprise-grade EdTech web application built specifically for the House of Edtech developer assignment. In strict adherence to assignment guidelines, the platform avoids basic CRUD concepts (such as generic to-do lists or basic task managers) and instead delivers a sophisticated, high-impact learning ecosystem featuring artificial intelligence tutoring, dynamic quiz generation, automated code rubric scoring, and role-based access control.

2. Requirements Compliance Matrix

Requirement Category	Assignment Criteria	EduPulse AI Implementation Status
Framework & Stack	Next.js 16 (TypeScript), React 19	App Router, Server Components, Streaming UI
Database Persistence	PostgreSQL / MongoDB / SQLite	Prisma ORM with SQLite (Local) & PostgreSQL (Prod)
CRUD & Beyond	Non-basic CRUD domain logic	Courses, Modules, Quizzes, Submissions & Analytics
Artificial Intelligence	AI SDK, OpenAI, Gemini or Groq	Google Gemini API + Heuristic Fallback Engine
Security & Auth	JWT Auth, RBAC, Data Validation	HttpOnly Cookie JWT, Bcrypt, Zod, Middleware RBAC
Testing & Quality	Unit & Integration Tests	Vitest Automated Test Suite (8/8 tests passing)
CI/CD & Deploy	GitHub Actions & Vercel Config	GitHub Actions CI pipeline (.github/workflows/ci.yml)

3. Technology Stack & Next.js 16 Architectural Decisions

- **Next.js 16 App Router:** Utilizes React 19 Server Components (RSC) to minimize client JavaScript bundle size and stream rendered HTML directly from the edge server.
- **Prisma ORM:** Provides compile-time type safety across database queries, declarative schema migrations, and seamless database portability between SQLite and PostgreSQL.
- **Tailwind CSS & Glassmorphism UI:** Custom design system incorporating modern dark mode palettes, vibrant color accents, micro-animations, and responsive layout grids.
- **Zod Payload Validation:** Enforces strict runtime validation on all REST API endpoints to sanitize user input and prevent injection or schema mismatch vulnerabilities.

4. Database Schema & Relational Model

The application relies on seven core database entities defined in `prisma/schema.prisma`:

- **User:** Stores user identities, hashed passwords (bcrypt), and roles (STUDENT, INSTRUCTOR, ADMIN).
- **Course:** Contains course titles, slugs, descriptions, difficulty levels, categories, and instructor relations.
- **Lesson:** Stores educational module content, reading duration, and ordering sequence.
- **Quiz & Question:** Houses multi-choice assessment questions, options array JSON, correct answer index, and explanations.
- **QuizAttempt:** Tracks student scores, completion timestamps, total questions, and chosen options.
- **Submission:** Records student practice code submissions alongside AI-evaluated scores and constructive feedback.

5. Security & Authentication Model

- **JWT HttpOnly Cookies:** Authentication tokens are signed using HS256 algorithm via jose library and transmitted exclusively inside HttpOnly, SameSite=Lax cookies to mitigate XSS attacks.
- **Edge Middleware RBAC:** Next.js Edge Middleware intercepts protected route requests (`/dashboard`, `/instructor`, `/api/courses`) to verify token validity and enforce role privileges.
- **Bcrypt Password Hashing:** Passwords are salted and hashed with a work factor of 10 prior to database insertion.

6. Artificial Intelligence Capabilities

EduPulse AI features an advanced multi-tier AI engine integrating Google Generative AI (@google/generative-ai, Gemini 1.5 Flash) with an intelligent heuristic fallback mechanism. This guarantees zero-downtime operation even when no external API key is provided.

- **Interactive AI Tutor Drawer:** Students can open a contextual chat drawer during any lesson to ask questions, request code examples, or receive simplified concept explanations.
- **AI Quiz Generator Studio:** Instructors can input any technical topic to automatically generate multiple-choice questions complete with answer keys and explanations.
- **AI Practice Code Evaluator:** Evaluates student code submissions against structured rubrics, assigning a numerical score (0-100), detailed feedback, and improvement recommendations.

7. Automated Testing Suite & CI/CD Pipeline

Comprehensive automated unit and integration tests are executed using Vitest:

- **Auth Test Suite (src/__tests__/auth.test.ts):** Verifies bcrypt password hashing, token generation, and JWT signature verification.
 - **Validation Test Suite (src/__tests__/validations.test.ts):** Tests boundary validation rules for registration, login, and course creation payloads.
 - **AI Module Test Suite (src/__tests__/ai.test.ts):** Ensures tutor responses, quiz question parsers, and submission graders return valid data structures.
- Test Suite Verification Result: 8 of 8 tests passed successfully across all modules.**

8. CI/CD Workflow

A GitHub Actions pipeline (.github/workflows/ci.yml) triggers on every repository commit to install dependencies, run Prisma generation, execute the Vitest suite, and verify Next.js production compilation.

9. Step-by-Step Guide to Build & Run

- Step 1: Environment Initialization — Created package.json with Next.js 16, React 19, TypeScript, Tailwind CSS, Prisma, and Vitest.
- Step 2: Database Schema & Seeding — Defined Prisma models in prisma/schema.prisma and populated demo courses/users via prisma/seed.ts.
- Step 3: Security & Middleware — Configured JWT cookie handlers in src/lib/auth.ts and Edge RBAC protection in src/middleware.ts.
- Step 4: AI Services Implementation — Built Google Gemini integration with heuristic fallback logic in src/lib/ai.ts.
- Step 5: API & REST Handlers — Developed App Router endpoints for auth, courses, quiz evaluation, and AI features.
- Step 6: UI Component & Pages Assembly — Built landing page, student dashboard, course reader, quiz player, and instructor studio.
- Step 7: Verification & Testing — Ran Vitest test suite and verified 100% test pass rate.
- Step 8: Automated PDF Generation — Created scripts/generate-pdf.js to build project documentation.

10. Candidate Profile & Submission Details

Candidate Name: [Sukruti Panigrahi](#)

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